

# Apply filters to SQL queries

## Project description

### **Describe what you accomplish through SQL**

As a security professional at a large organization, I have been tasked to inspect security issues to help keep the system secure. To examine the data, two tables “employees” and “log\_in\_attempts”. Then to implement filter operators like AND, NOT, OR, LIKE to look for any suspected failed login attempts.

## Retrieve after hours failed login attempts

### **Retrieve login attempts on specific dates**

In this case, by going through tables “employees” and “log\_in\_attempts”, filter using “>” 18:00 which is 6pm when office ends, and filter using these operators to look for any failed login attempts after 18:00. For failed attempts use “0” as False, and “1” as True, using boolean logic.

Command for SQL:

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00' AND success = 0;
```

## Retrieve login attempts on specific dates

Here, from table “log\_in\_attempts” filter out the login attempts between dates: “2022-05-08” and “2022-05-09”

```
SELECT *  
FROM log_in_attempts  
WHERE login_date = '2022-05-08' AND login_date = '2022-05-09';
```

## Retrieve login attempts outside of Mexico

Here, within the table, now NOT including Mexico as country to filter out all “Mexico” term used and show or output rest of the countries

```
SELECT *  
FROM log_in_attempts  
WHERE NOT country LIKE 'MEX%';
```

## Retrieve employees in Marketing

Here, we now select the employees table and filter out all department with “Marketing” and filter out “East” offices, using operator LIKE.

```
SELECT *  
FROM employees  
WHERE department = 'Marketing' AND office LIKE 'East%';
```

## Retrieve employees in Finance or Sales

Here, we filter out department with both Sales and Finance within same column together from employees table

```
SELECT *  
FROM employees  
WHERE department = 'Sales' AND department = 'Finance';
```

## Retrieve all employees not in IT

Here, filter out all the departments that's not “IT” and output the rest of the department.

```
SELECT *  
FROM employees  
WHERE NOT department LIKE 'Information Technology%';
```

## Summary

Overall, to identify unusual login attempts, first we filter through any failed login attempts after 6pm, and then filter out different dates, countries, looking for specific departments as well. If certain names are not clear while filtering through data, use “%” to get a similar name like Mexico for “Mex%”. For AND and OR, use conditions that include both data, and OR for either one is included. For NOT, it's excluding specific data, but output the rest of the data. LIKE is for any similarities with specific terms like “Mex%” that returns all related to Mexico.